Bonus Assignment (SVM with Kernels)

- Develop your own SVM library (from Scratch)
- You may use any Quadratic Programming solver (e.g. quadprog, CVXOPT, ..) to solve the Lagrangian Equation.
- Make use of Kernel functions (e.g. Quadratic , Polynomial, or RBF)
- For each of the given dataset (2-D):
  - Draw the dataset points using color coding (assign same color to points belong to same class)
  - Divide the background into grid, classify each point in the grid, give each point same color code of the class it belongs to.